

**LAND APPLICATION
POLLUTANT METALS CONCENTRATIONS**

Facility Name:

Cadillac WWTF

NPDES Permit No.:

Year: 2020

Pollutant	Celling Concentration	Time Period From: 11/19 To: 11/20	Time Period From: 4/30 To: 5/4	Time Period From: To:	Time Period From: To:	Time Period From: To:
	mg/kg dry basis	Concentration in Sewage Sludge, mg/kg dry basis	Concentration in Sewage Sludge, mg/kg dry basis	Concentration in Sewage Sludge, mg/kg dry basis	Concentration in Sewage Sludge, mg/kg dry basis	Concentration in Sewage Sludge, mg/kg dry basis
% Solids						
Arsenic	75	19.60	15.10			
Cadmium	85	1.70	1.50			
Copper	4,300	903.00	868.00			
Lead	840	30.10	20.40			
Mercury	57	1.70	6.00			
Molybdenum	75	26.50	23.80			
Nickel	420	24.80	26.60			
Selenium	100	4.90	5.10			
Zinc	7,500	1,410.00	1,140.00			

Heavy Metals Calculations

If the result is less than the analytical detection limits enter the less than sign (<) followed by the minimum reportable limit but DO NOT enter acronym's such as, "ND", "BQL", "BMRL", etc.

Pollutant	10/1/13	10/2/13	10/3/13	10/4/13	10/5/13	10/6/13	10/7/13	10/8/13	10/9/13	10/10/13	10/11/13	10/12/13	10/13/13	10/14/13	10/15/13	10/16/13	10/17/13	10/18/13	10/19/13	10/20/13	10/21/13	10/22/13	10/23/13	10/24/13	10/25/13	10/26/13	10/27/13	10/28/13	10/29/13	10/30/13	10/31/13	11/1/13	11/2/13	11/3/13	11/4/13	11/5/13	11/6/13	11/7/13	11/8/13	11/9/13	11/10/13	11/11/13	11/12/13	11/13/13	11/14/13	11/15/13	11/16/13	11/17/13	11/18/13	11/19/13	11/20/13	11/21/13	11/22/13	11/23/13	11/24/13	11/25/13	11/26/13	11/27/13	11/28/13	11/29/13	11/30/13	12/1/13	12/2/13	12/3/13	12/4/13	12/5/13	12/6/13	12/7/13	12/8/13	12/9/13	12/10/13	12/11/13	12/12/13	12/13/13	12/14/13	12/15/13	12/16/13	12/17/13	12/18/13	12/19/13	12/20/13	12/21/13	12/22/13	12/23/13	12/24/13	12/25/13	12/26/13	12/27/13	12/28/13	12/29/13	12/30/13	12/31/13	1/1/14	1/2/14	1/3/14	1/4/14	1/5/14	1/6/14	1/7/14	1/8/14	1/9/14	1/10/14	1/11/14	1/12/14	1/13/14	1/14/14	1/15/14	1/16/14	1/17/14	1/18/14	1/19/14	1/20/14	1/21/14	1/22/14	1/23/14	1/24/14	1/25/14	1/26/14	1/27/14	1/28/14	1/29/14	1/30/14	1/31/14	2/1/14	2/2/14	2/3/14	2/4/14	2/5/14	2/6/14	2/7/14	2/8/14	2/9/14	2/10/14	2/11/14	2/12/14	2/13/14	2/14/14	2/15/14	2/16/14	2/17/14	2/18/14	2/19/14	2/20/14	2/21/14	2/22/14	2/23/14	2/24/14	2/25/14	2/26/14	2/27/14	2/28/14	2/29/14	2/30/14	3/1/14	3/2/14	3/3/14	3/4/14	3/5/14	3/6/14	3/7/14	3/8/14	3/9/14	3/10/14	3/11/14	3/12/14	3/13/14	3/14/14	3/15/14	3/16/14	3/17/14	3/18/14	3/19/14	3/20/14	3/21/14	3/22/14	3/23/14	3/24/14	3/25/14	3/26/14	3/27/14	3/28/14	3/29/14	3/30/14	3/31/14	4/1/14	4/2/14	4/3/14	4/4/14	4/5/14	4/6/14	4/7/14	4/8/14	4/9/14	4/10/14	4/11/14	4/12/14	4/13/14	4/14/14	4/15/14	4/16/14	4/17/14	4/18/14	4/19/14	4/20/14	4/21/14	4/22/14	4/23/14	4/24/14	4/25/14	4/26/14	4/27/14	4/28/14	4/29/14	4/30/14	5/1/14	5/2/14	5/3/14	5/4/14	5/5/14	5/6/14	5/7/14	5/8/14	5/9/14	5/10/14	5/11/14	5/12/14	5/13/14	5/14/14	5/15/14	5/16/14	5/17/14	5/18/14	5/19/14	5/20/14	5/21/14	5/22/14	5/23/14	5/24/14	5/25/14	5/26/14	5/27/14	5/28/14	5/29/14	5/30/14	5/31/14	6/1/14	6/2/14	6/3/14	6/4/14	6/5/14	6/6/14	6/7/14	6/8/14	6/9/14	6/10/14	6/11/14	6/12/14	6/13/14	6/14/14	6/15/14	6/16/14	6/17/14	6/18/14	6/19/14	6/20/14	6/21/14	6/22/14	6/23/14	6/24/14	6/25/14	6/26/14	6/27/14	6/28/14	6/29/14	6/30/14	7/1/14	7/2/14	7/3/14	7/4/14	7/5/14	7/6/14	7/7/14	7/8/14	7/9/14	7/10/14	7/11/14	7/12/14	7/13/14	7/14/14	7/15/14	7/16/14	7/17/14	7/18/14	7/19/14	7/20/14	7/21/14	7/22/14	7/23/14	7/24/14	7/25/14	7/26/14	7/27/14	7/28/14	7/29/14	7/30/14	7/31/14	8/1/14	8/2/14	8/3/14	8/4/14	8/5/14	8/6/14	8/7/14	8/8/14	8/9/14	8/10/14	8/11/14	8/12/14	8/13/14	8/14/14	8/15/14	8/16/14	8/17/14	8/18/14	8/19/14	8/20/14	8/21/14	8/22/14	8/23/14	8/24/14	8/25/14	8/26/14	8/27/14	8/28/14	8/29/14	8/30/14	8/31/14	9/1/14	9/2/14	9/3/14	9/4/14	9/5/14	9/6/14	9/7/14	9/8/14	9/9/14	9/10/14	9/11/14	9/12/14	9/13/14	9/14/14	9/15/14	9/16/14	9/17/14	9/18/14	9/19/14	9/20/14	9/21/14	9/22/14	9/23/14	9/24/14	9/25/14	9/26/14	9/27/14	9/28/14	9/29/14	9/30/14	10/1/14	10/2/14	10/3/14	10/4/14	10/5/14	10/6/14	10/7/14	10/8/14	10/9/14	10/10/14	10/11/14	10/12/14	10/13/14	10/14/14	10/15/14	10/16/14	10/17/14	10/18/14	10/19/14	10/20/14	10/21/14	10/22/14	10/23/14	10/24/14	10/25/14	10/26/14	10/27/14	10/28/14	10/29/14	10/30/14	10/31/14	11/1/14	11/2/14	11/3/14	11/4/14	11/5/14	11/6/14	11/7/14	11/8/14	11/9/14	11/10/14	11/11/14	11/12/14	11/13/14	11/14/14	11/15/14	11/16/14	11/17/14	11/18/14	11/19/14	11/20/14	11/21/14	11/22/14	11/23/14	11/24/14	11/25/14	11/26/14	11/27/14	11/28/14	11/29/14	11/30/14	12/1/14	12/2/14	12/3/14	12/4/14	12/5/14	12/6/14	12/7/14	12/8/14	12/9/14	12/10/14	12/11/14	12/12/14	12/13/14	12/14/14	12/15/14	12/16/14	12/17/14	12/18/14	12/19/14	12/20/14	12/21/14	12/22/14	12/23/14	12/24/14	12/25/14	12/26/14	12/27/14	12/28/14	12/29/14	12/30/14	12/31/14																																																																																																												
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For Calculating the Agronomic Rate for the Land Application of Sewage Sludge

From analysis of the sewage sludge to be land applied, determine the amount of each nitrogen compound, based on dry weight, in pounds per ton (lbs/ton).

Nitrogen compound	Concentration of Nitrogen Compounds							
	(mg/kg)	(mg/kg)	(mg/kg)	(mg/kg)	(mg/kg)	(mg/kg)	(mg/kg)	(mg/kg)
Total Kjeldahl Nitrogen (TKN as N)	71,500	71,500	71,500	71,500	73,900	76,900		
Ammonium Nitrogen (NH ₄ as N)	42,100	42,100	42,100	42,100	29,200	29,200		
Nitrate Nitrogen (NO ₃ as N)	214	214	214	214	145	145		
Total Phosphorus (mg/kg or mg/L)	52,000	52,000	52,000	52,000	40,400	40,400		

Step 2:

Assume that the available traction (Kv) is dependent upon operations at the site (see Table 1). Use the following table and enter the appropriate Kv Factor

If Sewage Sludge is:	Factor K_1 Is:
Liquid and Surface Applied	0.25
Liquid and Incorporated into the Soil	1.00
Landfilled and Applied in any Manner	1.00

Factor K_g is:[illegible]

Step 3:

Calculate the amount of organic nitrogen available in the sewage sludge to be applied. The factor f , used for determining the amount of Org as N present due to mineralization is provided below in Table 2. The value of F is dependent upon how the sludge is treated (i.e., aerobic digestion, composted, etc.).

[illegible]

Step 4

Calculation is determined on data entered in Steps 1 through Steps 3

Step 5:

Nitrogen Credits The available nitrogen credits from previous legume crops and existing residuals must be determined under there will be a previous crop nitrogen credit in the soil. Select the appropriate nitrogen credit based upon the Step 5a. Previous Legume Crop, Nprevious crop. If the crop for the previous year was a legume and was plowed data shown in Table 3.

Table 3. Nitrogen Credits from Legumes in Rotations	
Legume Crop	Nitrogen Credit
Aralia (1st year after)	100-140 lbs/acre
>80% stand	60-100 lbs/acre
60-80% stand	0-60 lbs/acre
<60% stand	Half of 1st year credit
Aralia (2nd year after)	100-120 lbs/acre
Sweet Clover	40-80 lbs/acre
Red Clover	30-60 lbs/acre
Soybeans*	No credit for wheat double-cropped after soybean harvest
*Allow 1 pound of N credit per bushel of yield	

If you do not have a "nitrogen credit" please enter 0 for the appropriate site

	Site 1	Site 2	Site 3	Site 4	Site 5	Site 6	Site 7	Site 8	Site 9	Site 10
Nitrogen Credit	0	0	0	0	0	0	0			

Step 5b:

Existing Nitrate Content of Soil, N_{existing} The nitrogen credit for the existing nitrate level in the soil can be accounted for by using the soil test nitrate results in the following equations. (See soil testing procedures for soil sampling methods.)

	Site 1	Site 2	Site 3	Site 4	Site 5	Site 6	Site 7	Site 8	Site 9	Site 10
Depth of Sample (inches)	5	6	6	6	6	6	6			
$\text{NO}_3\text{ as } N_{\text{ext}}$	0	0	0	0	0	0	0			

Step 5c:

Previous Sludge, Available Organic Nitrogen, Previous Org(N)available Because the mineralization of organic nitrogen in sludge occurs over a long time, there will be a nitrogen credit for mineralization of previously applied sludge. This nitrogen credit is 0.5 of the previous year's calculated total available organic nitrogen.

Abbreviations	Definition	Location of Information
PYCONA	Previous Year Current Organic (lbs/ton) $N_{\text{available}}$	Step 3 of Last Years Report
PYSAR	Previous Years Sludge Application Rate (Dry Tons / Acre)	Last Year Data Entry - General Information Section
PYCONA		
PYSAR		

Step 5D:

Calculation is determined on data entered in Steps 5A, Step 5B and Step 5C.

Step 6:

Determine the crop nitrogen requirements

Table 4.	
CROP	CROP FACTOR
Wheat	1.75 lbs Nitrogen
Corn	1.35 lbs Nitrogen
Oats	1.15 lbs Nitrogen
Grain Sorghum/Milo	1.35 lbs Nitrogen
Barley	1.50 lbs Nitrogen
Soybeans	5.4 lbs Nitrogen
Alfalfa	56 lbs Nitrogen
Orchard Grass	50 lbs Nitrogen
Brome Grass	33 lbs Nitrogen
Sudangrass	50 lbs N/1000 lbs seed
Tall Fescue	39 lbs Nitrogen
Perennial Sorghum	9 lbs Nitrogen

Table 5. Soil Factor	
Soil Type	Factor
Sandy	1.1
All Other	1

	Site 1	Site 2	Site 3	Site 4	Site 5	Site 6	Site 7	Site 8	Site 9	Site 10
Select Crop (Table 4)	Corn	Corn	Corn	Corn	Corn	Corn	Corn			
Crop Factor (Table 4)	1.35	1.35	1.35	1.35	1.35	1.35	1.35			
Crop Yield	100	100	100	100	100	100	100			
Soil Type (Table 5)	1	1	1	1	1	1	1			
Soil Factor (Table 5)	1	1	1	1	1	1	1			

Step 7:

Calculation is determined on data entered in Steps 5D and Step 6.

Step 8:

Calculation is determined on data entered in Steps 7 and Step 4.

Step 9:

Calculation is determined on data entered in Step 8.

Step 10:

From the soil analysis record the following parameters

Site Number	0 to 6 inches			Nitrate as Nitrogen
	Available Phosphorus (mg/L)	Exchangeable Potassium (mg/L)	pH (SU)	
Site No. 1	40	109	6.1	0
Site No. 2	22	78	6.8	0
Site No. 3	40	155	6.7	0
Site No. 4	47	97	6.5	0
Site No. 5	140	62	6.5	0
Site No. 6	65	97	7	0
Site No. 7	129	180	7.1	0
Site No. 8				
Site No. 9				
Site No. 10				